

PYTHON CLASS PRACTISE CODE

14/07/26 – PRACTICE PROGRAMS

1.

The screenshot shows the SIMATS Online Programming Lab interface. The header is dark blue with the text "SIMATS Online Programming Lab" and "PYTHON PROGRAMMING" below it. A user profile "Sivarangjenii B" with ID "192521375RD" is in the top right. The main area is divided into three sections: "PROGRAM EDITOR" on the left, a central control area, and "INPUT" on the right. The "PROGRAM EDITOR" section contains a Python code snippet. The central control area has a language dropdown set to "Python", a green "Run" button, and a blue "Save" button. The "INPUT" section shows the number "29".

```
1 #positive number
2 a = int(input("enter a number: "))
3 if(a>=0):
4     print("positive number");
```

OUTPUT

```
enter a number: positive number
```

2.

The screenshot shows the SIMATS Online Programming Lab interface for a second program. The header and user profile are identical to the first screenshot. The "PROGRAM EDITOR" section contains a different Python code snippet. The central control area has the same "Python" dropdown, "Run" button, and "Save" button. The "INPUT" section shows the number "8000".

```
1 a = int(input("enter a number: "))
2 if(a>2000):
3     a=a-500
4     print("discount amount",a)
```

OUTPUT

```
enter a number: discount amount 7500
```

3.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 a = int(input("enter a number:"))
2 category = (input("enter a category: "))
3 if(category == "sports"):
4     bonus = 15
5     total= a+bonus
6     print("Total marks",total)
```

INPUT

79
sports

OUTPUT

```
enter a number:enter a category: Total marks 94
```

4.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 year=int(input("enter a year: "))
2 if(year % 400 == 0):
3     print("leap year")
4 else:
5     print("not a leap year")
```

INPUT

2007

OUTPUT

```
enter a year: not a leap year
```

5.

SIMATS Online Programming Lab
PYTHON PROGRAMMING

 Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 age=int(input("enter a number: "))
2 if(age>=18):
3     print("eligible")
4 else:
5     print("not eligible")
```

INPUT

17

OUTPUT

```
enter a number: not eligible
```

6.

SIMATS Online Programming Lab
PYTHON PROGRAMMING

 Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #to check whether given number is 0 or not
2 a=int(input("enter a number: "))
3 if(a==0):
4     print("zero")
5 else:
6     print("not a zero")
```

INPUT

9

OUTPUT

```
enter a number: not a zero
```

7.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #check positive or negative number
2 a=int(input("enter a number: "))
3 if(a>0):
4     print("positive number")
5 elif a==0:
6     print("zero")
7 else:
8     print("negative number")
```

INPUT

91

OUTPUT

```
enter a number: positive number
```

8.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #largest of 3 numbers
2 a=int(input("enter first number: ""\n"))
3 b=int(input("enter second number: ""\n"))
4 c=int(input("enter third number: ""\n"))
5 if(a>b):
6     print("largest number is a")
7 elif a<b:
8     print("largest number is b")
9 else:
10    print("largest number is c")
```

INPUT

21
34
6

OUTPUT

```
enter first number:
enter second number:
enter third number:
largest number is b
```

9.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B

192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #2digit number
2 a=int(input("enter a number: "))
3 if(a>9):
4     print("two digit number")
5 elif(a>99):
6     print("three digit number")
7 else:
8     print("not a two digit number")
```

INPUT

89

OUTPUT

```
enter a number: two digit number
```

10.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B

192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #divisble by 3or5
2 a = int(input("enter a number: "))
3 if(a%3==0):
4     print("divisible by 3")
5 elif a%5==0:
6     print("divisible by 5")
7 else:
8     print ("not divisible")
```

INPUT

9

OUTPUT

```
enter a number: divisible by 3
```

11.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B

192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #5or10
2 num = int(input("enter a number "))
3 if num%5==0 or num%10==0:
4     print("the no is divisible by 5 or 10")
5 else:
6     print("the no is not divisble by 5 or 10")
```

INPUT

24

OUTPUT

```
enter a number the no is not divisble by 5 or 10
```

12.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B

192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #single,digit,above 2 digit,negative
2 num = int(input("enter a no: "))
3 if num<0:
4     print("negative no")
5 elif num>=0 and num<=9:
6     print("single digit")
7 elif num>= 10 and num <=99:
8     print("above two digits")
9 else:
10    print("above two digits")
```

INPUT

88

OUTPUT

```
enter a no: above two digits
```

13.

The screenshot shows the SIMATS Online Programming Lab interface. The header is dark blue with the text "SIMATS Online Programming Lab" and "PYTHON PROGRAMMING" below it. On the right, a user profile for "Sivarangjenii B" with ID "192521375RD" is visible. The main area is divided into two sections: "PROGRAM EDITOR" on the left and "INPUT" on the right. The "PROGRAM EDITOR" section has a dropdown menu set to "Python" and buttons for "Run" and "Save". The code editor contains the following Python code:

```
1 #divisble by 5 or not
2 num = int(input("enter a no: "))
3 if num%5==0:
4     print("the number is divisible by 5")
5 else:
6     print("not divisible by 5")
7     print("remainder =",num%5)
```

The "INPUT" section shows the number "38" entered.

OUTPUT

```
enter a no: not divisible by 5
remainder = 3
```

14.

The screenshot shows the SIMATS Online Programming Lab interface. The header is dark blue with the text "SIMATS Online Programming Lab" and "PYTHON PROGRAMMING" below it. On the right, a user profile for "Sivarangjenii B" with ID "192521375RD" is visible. The main area is divided into two sections: "PROGRAM EDITOR" on the left and "INPUT" on the right. The "PROGRAM EDITOR" section has a dropdown menu set to "Python" and buttons for "Run" and "Save". The code editor contains the following Python code:

```
1 #student mark system
2 mark = int(input("enter the mark: "))
3 if mark>=90:
4     print("grade A")
5 elif mark>=75:
6     print("Gradde B")
7 elif mark>=50:
8     print("Grade C")
9 else:
10    print("fail")
```

The "INPUT" section shows the number "97" entered.

OUTPUT

```
enter the mark: grade A
```

15.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #traffic light system
2 color=input("enter the traffic light: ")
3 if color == "red":
4     print("stop")
5 elif color == "yellow":
6     print("wait")
7 elif color == "green":
8     print("go")
9 else:
10    print("invalid color")
```

INPUT

red

OUTPUT

```
enter the traffic light: stop
```

16.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #roots of quadratic eqn
2 a=int(input("enter a: "))
3 b=int(input("enter b: "))
4 c=int(input("enter c: "))
5 d=b*b-4*a*c
6 if d>0:
7     print("two real roots")
8 elif d==0:
9     print("one real root")
10 else:
11    print("imaginary roots")
```

INPUT

2
4
6

OUTPUT

```
enter a: enter b: enter c: imaginary roots
```


17.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #evn or odd
2 num = int(input("enter a number: "))
3 if num%2==0:
4     print("even")
5 else:
6     print("odd")
```

INPUT

99

OUTPUT

enter a number: odd

18.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #age category
2 age=int(input("enter age: "))
3 if age<13:
4     print("child")
5 elif age<20:
6     print("teenagers")
7 elif age<60:
8     print("adult")
9 else:
10    print("senior adults")
```

INPUT

100

OUTPUT

enter age: senior adults

WHILE LOOP

19.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #10 to 1
2 i = 10
3 while(i>=1):
4     print(i,end=" ")
5     i = i-1
```

INPUT

Your INPUT goes here!

OUTPUT

10 9 8 7 6 5 4 3 2 1

20.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #1 to 10
2 i = 1
3 while(i<=10):
4     print(i)
5     i = i+1
```

INPUT

Your INPUT goes here!

OUTPUT

1
2
3
4
5
6
7
8
9
10

21.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B

192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #odd numbers
2 i=1
3 while(i<=20):
4     if(i%2 !=0):
5         print(i)
6     i=i+1
```

INPUT

Your INPUT goes here!

OUTPUT

```
1
3
5
7
9
11
13
15
17
19
```

22.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

 Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #fibonacci series
2 n=int(input("enter a number: "))
3 a=0
4 b=1
5 count=0
6 print("fibonacci series: ")
7 while(count<n):
8     print(a,end=" ")
9     c=a+b
10    a=b
11    b=c
12    count = count+1
```

INPUT

9

OUTPUT

```
enter a number: fibonacci series:
0 1 1 2 3 5 8 13 21
```

23.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

 Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #reverse the number
2 n=int(input("enter a number: "))
3 rev = 0
4 while(n>0):
5     a=n%10
6     rev=rev*10+a
7     n=n//10
8     print(rev)
```

INPUT

8743

OUTPUT

```
enter a number: 3
34
347
3478
```

24.

SIMATS Online Programming Lab
PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #sum of series
2 n=int(input("enter a number: "))
3 i=0
4 sum = 0
5 while(i<=5):
6     sum = sum+i
7     i=i+1
8     print(sum)
```

INPUT

5

OUTPUT

```
enter a number: 0
1
3
6
10
15
```

25.

SIMATS Online Programming Lab
PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #sum of even no
2 n=int(input("enter a number: "))
3 i=2
4 sum=0
5 while(i<=n):
6     sum=sum+i
7     i=i+2
8     print("sum of even numbers=",sum)
```

INPUT

10

OUTPUT

```
enter a number: sum of even numbers= 2
sum of even numbers= 6
sum of even numbers= 12
sum of even numbers= 20
sum of even numbers= 30
```

26.

SIMATS Online Programming Lab
PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #factorial of number
2 num=int(input("enter a number: "))
3 fact =1
4 i=1
5 while(i<=num):
6     fact = fact*i
7     i=i+1
8     print("factorial=",fact)
```

INPUT

5

OUTPUT

```
enter a number: factorial= 1
factorial= 2
factorial= 6
factorial= 24
factorial= 120
```

27.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #sum of even fibonacci series
2 n = int(input("enter the limit: "))
3 a = 0
4 b = 1
5 sum = 0
6 while a<=n:
7     if a%2 == 0:
8         sum = sum + a
9     c = a + b
10    a = b
11    b = c
12    print("sum of even fibonacci numbers =",sum)
```

INPUT

10

OUTPUT

enter the limit: sum of even fibonacci numbers =
sum of even fibonacci numbers = 0
sum of even fibonacci numbers = 0
sum of even fibonacci numbers = 2
sum of even fibonacci numbers = 2
sum of even fibonacci numbers = 2
sum of even fibonacci numbers = 10

OUTPUT

```
enter the limit: sum of even fibonacci numbers = 0
sum of even fibonacci numbers = 0
sum of even fibonacci numbers = 0
sum of even fibonacci numbers = 2
sum of even fibonacci numbers = 2
sum of even fibonacci numbers = 2
sum of even fibonacci numbers = 10
```

28.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #sum of digits
2 n = int(input("enter a number: "))
3 sum = 0
4 while n>0:
5     digit = n%10
6     sum = sum+digit
7     n = n//10
8     print("sum of digits =",sum)
```

INPUT

20

OUTPUT

```
enter a number: sum of digits = 0
sum of digits = 2
```

29.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #armstrong number
2 n = int(input("enter a number: "))
3 temp = n
4 sum = 0
5 while temp > 0:
6     digit = temp%10
7     sum = sum+digit**3
8     temp = temp//10
9     if sum == n:
10         print("armstrong no")
11     else:
12         print("not an armstrong no")
```

INPUT

23

OUTPUT

```
enter a number: not an armstrong no
not an armstrong no
```


30.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B

192521375RD

PROGRAM EDITOR

Python

Run

Save

```
1 #palindrome number
2 n = int(input("enter a number: "))
3 temp = n
4 reverse = 0
5 while temp>0:
6     digit = temp%10
7     reverse = reverse*10+digit
8     temp = temp//10
9     if reverse == n:
10         print("palindrome")
11     else:
12         print("not a palindrome")
```

INPUT

24

OUTPUT

```
enter a number: not a palindrome
not a palindrome
```